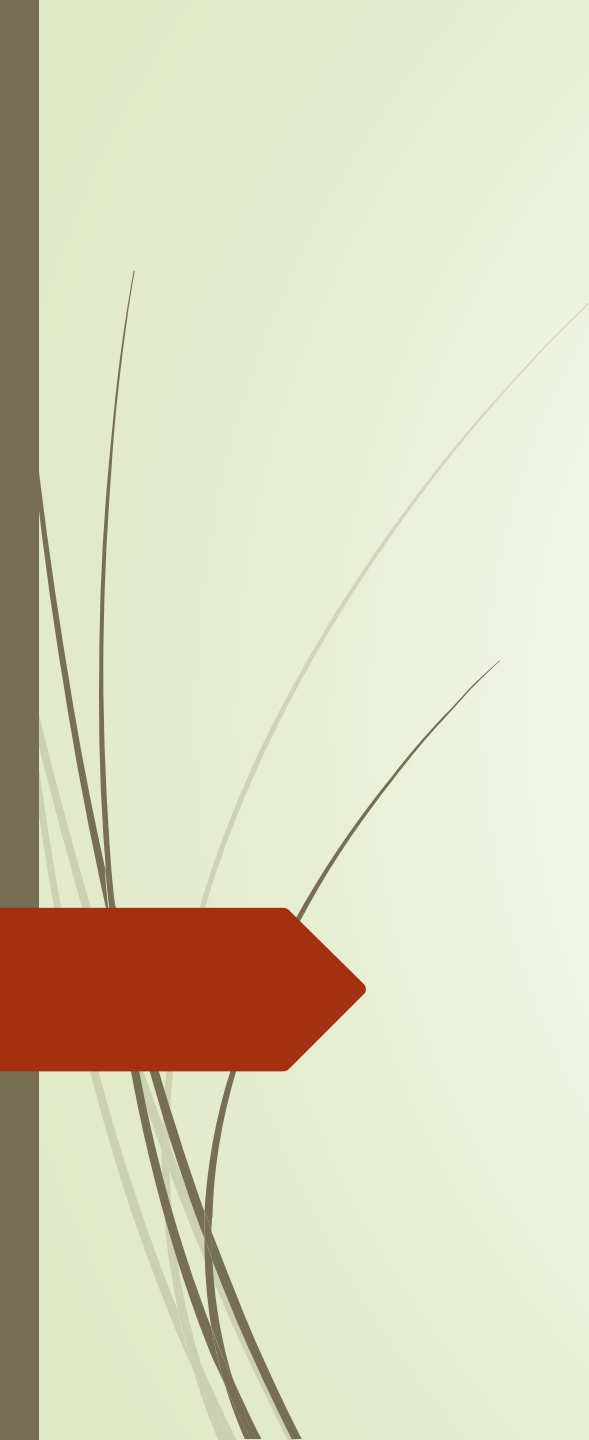




The Effects of Engineering Applications on Health, Environment and Security in Universal and Social Aspects



Introduction

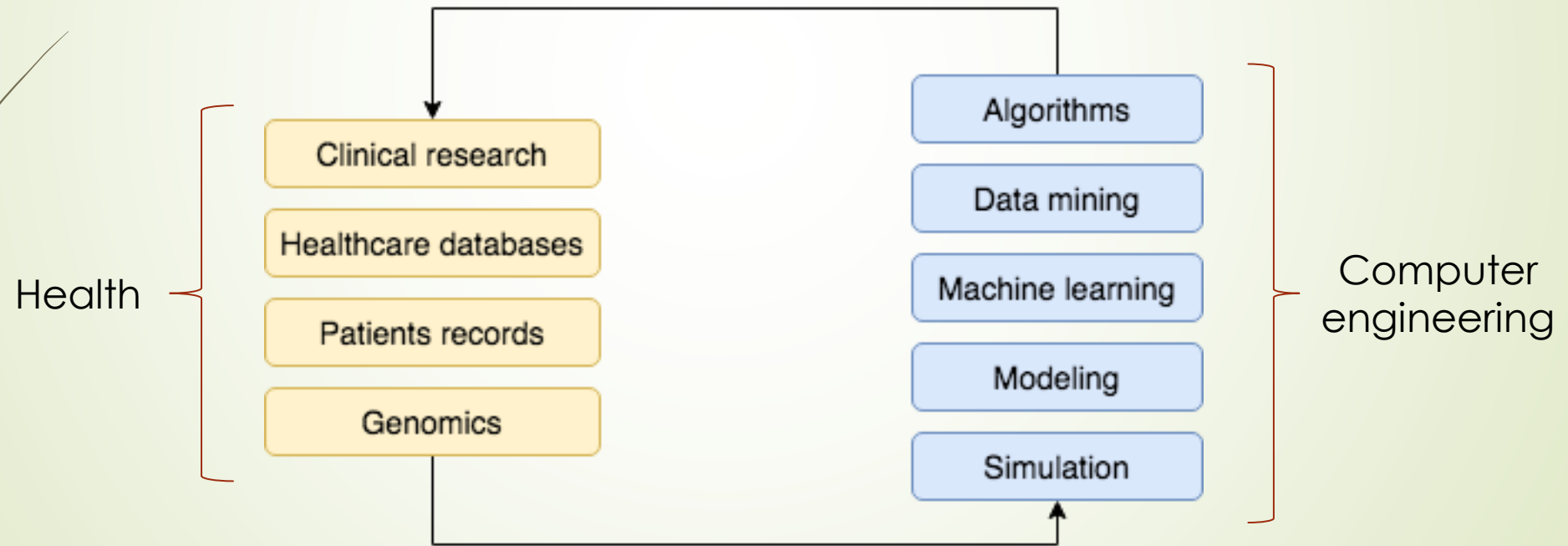
- Computer engineering is the discipline that embodies science and technology of design, implementation, and maintenance of software and hardware components.
- In this presentation, we discuss applications and effects of computer engineering in three main fields:
 1. Health
 2. Environment
 3. Security



Computer Engineering and Health

Computer Engineering and Healthcare

- Computer engineering revolutionizes the healthcare industry in many aspects.



Recent Advances in Healthcare Technology

- Recent advances in healthcare technology involving computer engineering:
 - Artificial intelligence
 - Medical image processing for breast cancer analysis
 - Fast and cheaper drug development
 - Healthcare wearables
 - Sensors to manage weight, stress levels, or sleeping quality
 - Robotics
 - Surgical robots, exoskeletons, and disinfection robots
 - 3-D printing
 - Nanotechnology

Medical Automation Systems

- Medical automation systems help society regularize and ease healthcare services.
 - Centralized data management,
 - Online appointments,
 - Maintenance of records,
 - Patient health monitoring, ...
- Successful examples of these automations are also in use in Turkey:
 - MHRs → Web and mobile applications for patients to make appointments for hospitals and doctors
 - HES → Mobile application to track COVID-19 for individuals



Healthier Individuals, Healthier Society

- ▶ Computer engineering helps humanity create a healthier society by:
 - ▶ Reducing healthcare costs
 - ▶ Predicting epidemics
 - ▶ Avoiding preventable deaths
 - ▶ Improving life quality
 - ▶ Developing new drugs and treatments
 - ▶ Developing cheaper drugs
 - ▶ Reducing healthcare waste

Job Opportunities in Healthcare Industry

- The health industry has recently emerged as an area of new job opportunities for computer engineers.
 - Analysts, data scientists, software developers, system engineers, ...
- Fields that a computer engineer should know about to increase her/his preferability in healthcare industry:
 - Computer science
 - Organizational theory
 - Information science
 - Management science
 - Cognitive science

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Computer Engineering and Environment



Environmental Problems in Today's World

- Our environment is inevitably and constantly changing.
- Society should be aware of what types of environmental problems our planet is facing:
 - Pollution,
 - Global warming,
 - Soil degradation,
 - Climate change,
 - Loss of biodiversity,
 - Waste disposal,
 - Natural resource depletion, ...



Role of Computers in Protecting the Environment

- Computers and computer-guided applications have been actively used to preserve and protect the environment.
- **Digitalization**
 - A key factor in the reduction of environmental impacts as energy-efficient technologies become more integrated in houses, business, and industry.
 - Some examples of environmental protection via digitalization:
 - Less paper consumption in education and business
 - Energy saving by smart household electrical appliances
 - Online platforms for shopping, bill payment, or reading news



Computer-guided Applications to Protect the Environment

- Several interesting computer-guided applications are employed to monitor and resolve environmental problems.
 - Cleaning up oceans by monitoring and identifying sources of plastic wastes
 - Wind power prediction to produce cheaper wind energy
 - Computer-aided analysis on big data to reduce urban energy consumption
 - Remote monitoring of wildlife to gather information about species
 - Gene sequencing to protect endangered species



Green Computing

- Green computing is the practice of using computer resources efficiently.
- Some green computing advices that we can follow to reduce our own energy consumption:
 - Put devices in "sleep" mode when they are not in use.
 - Turn off devices when possible.
 - Prefer energy-efficient computer components (displays, printers, or laptops).
 - Contribute to recycling.



Environmental Threats of Computer Technologies

- ▶ Although computer technologies provide significant benefits in environmental protection, they also bring some threats to be considered attentively.
- ▶ **Manufacturing**
 - ▶ The production of digital electronics (e.g., circuits, batteries, and displays) requires large amounts of raw materials and energy.
- ▶ **Electrical strain**
 - ▶ Billions of devices are running concurrently, which implies a huge electricity consumption.
- ▶ **Disposal**
 - ▶ Millions of devices are taken out of service every year. Disposal process of these devices results in toxic chemicals and harmful particles.



Computer Engineering and Security

Security in Computing Systems

- ▶ Computer engineering principles involves security issues of modern computing systems.
- ▶ **Computer security** is the protection of computer systems and information from harm, theft, and unauthorized use.
 - ▶ Keeps information protected
 - ▶ Prevents viruses and malware
 - ▶ Maintains system performance (allows programs to run safe and fast)
 - ▶ Protects overall health of systems

Types of Computer Security

- There are different types of computer security to protect information from malicious threats and attacks:
 - Application security
 - Threats: DoS attacks, SQL injection, data breaches, ...
 - Precautions: Firewalls, encryption, antivirus software, ...
 - Information security
 - CIA triad (confidentiality – integrity – availability)
 - Network security
 - Endpoint security
 - Internet security

Biometric Security

- An alternative to traditional password mechanism
 - Translates a person's characteristics into digital codes
- Biometric identifiers:
 - Fingerprints
 - Hand geometry
 - Facial recognition
 - Voice
 - Retinal patterns



User Privacy

- User privacy is the right of individuals that concerns storing, repurposing, and displaying information to others (third-parties such as people and companies).
- Companies (computer engineers as employees) are responsible to protect user privacy.
 - Privacy-preserving systems
- In Turkey, user privacy is protected and ensured by laws.
 - KVKK → The protection of pseronal data in line with the fundamental rights related with privacy and freedom



Why is Computer Security Important?

- The society is more technologically reliant than ever before.
- Cybercrimes have been steadily increasing since the last decade.
 - Examples: information theft, social engineering, phishing, ...
 - Consequences: economical, reputational, and regulatory costs
- As people get more involved in technology and the Internet, computer security becomes more important due to existence of sensitive data, personal information, and intellectual property.



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